Module 3.4
Constraint satisfaction

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Outline

- Constraint Satisfaction
- Example
- Inferences in CSP,
- CSP backtracking algorithm

Constraint satisfaction problems (CSPs)

- **Consist of three components**
  - **A set of** variables X_i
  - **A set of** Domain D_i
  - **A set of** Constraints C_i *Specifiying allowable conbination of values subsets of variables*
- **Our task is to assign a value to each variable so that we fulfill all the constraints.**

Constraint satisfaction problems (CSPs)

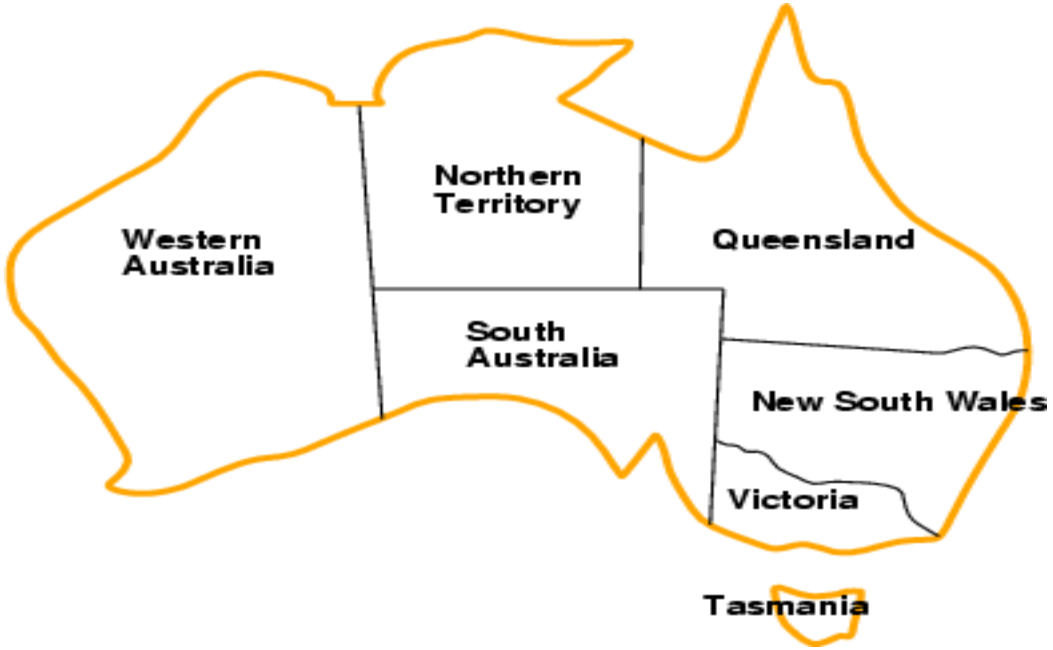
- It consists of the following:
  - A finite set of **variables** which stores the solution (V = {V1, V2, V3,....., Vn})
  - A set of **discrete** values known as **domain** from which the solution is picked (D = {D1, D2, D3,.....,Dn})
  - A finite set of **constraints** (C = {C1, C2, C3,....., Cn})

Uninformed ("Blind") Search Strategies

- No information is available other than
  - The current state
    - Its parent (perhaps complete path from initial state)
    - Its operators (to produce successors)
  - The goal test
  - The current path cost (cost from start state to current state)

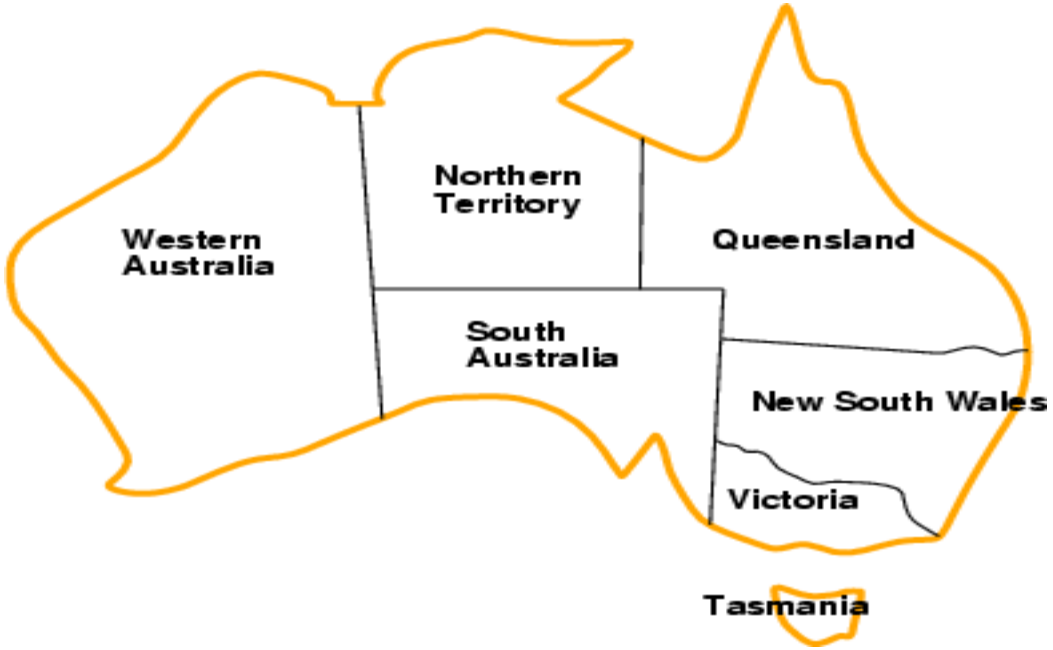
Example: Map-Coloring

Problem Statement: Colour each region either red, green or blue in such a way that no neighbouring regions have the same colour.



Example: Map-Coloring

- Variables *WA, NT, Q, NSW, V, SA, T*
- Domains D_i = {red, green, blue}
  - Constraints: adjacent regions must have different colors
  - e.g., WA ≠ NT, or (WA,NT) in {(red, green),(red, blue), (green, red), (green, blue), (blue, red), (blue, green)}



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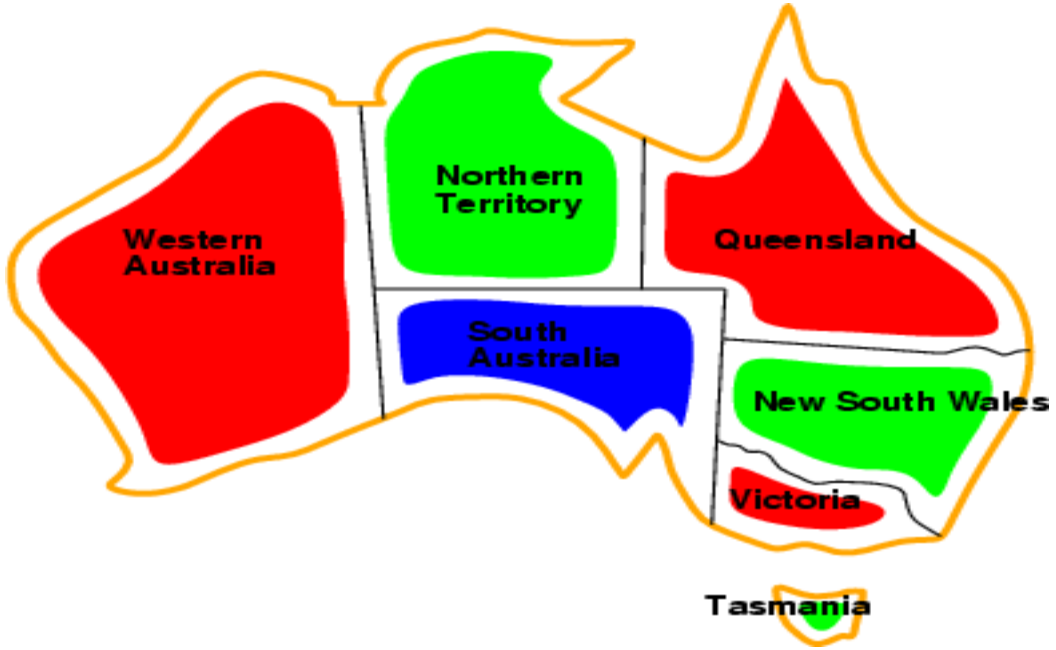
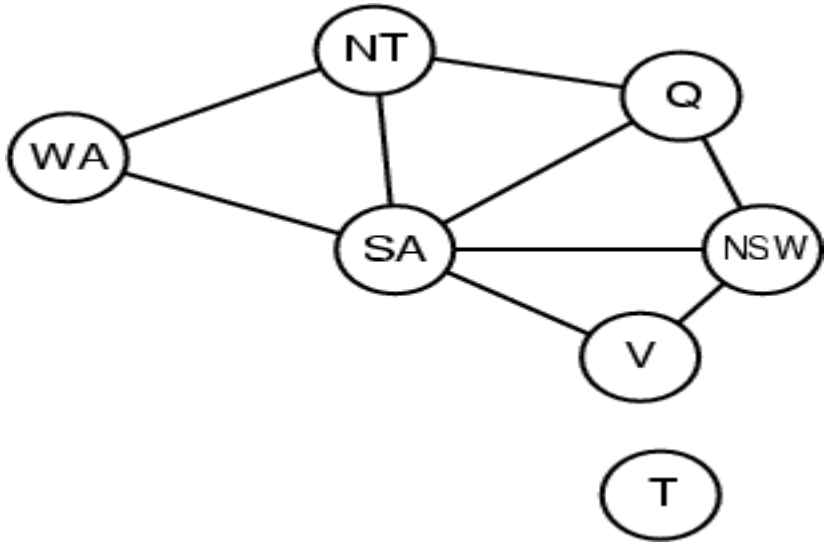


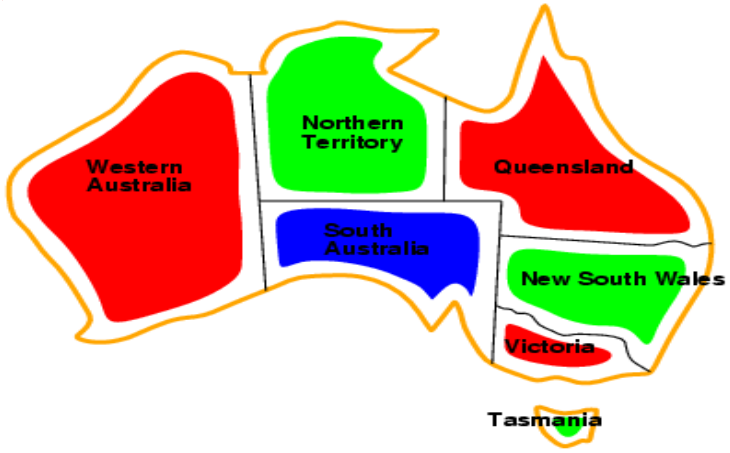
- Solutions are complete and consistent assignments,
  - e.g., WA = red, NT = green,Q = red,NSW = green,V = red,SA = blue,T = green

Constraint graph

- Binary CSP: each constraint relates two variables
- Constraint graph: nodes are variables, arcs are constraints



Varieties of constraints



- Unary constraints involve a single variable,
  - e.g., SA $\neq$ green
- Binary constraints involve pairs of variables,
  - e.g., SA $\neq$ WA
- Higher-order constraints involve 3 or more variables,
  - e.g., crypt arithmetic column constraints

Crypt - arithmetic Problems

- Also called as alphabetic/verbal arithmetic/word addition
- Mathematical game where a mathematical equation is presented where digits in the equation are replaced by letters.
- E.g. **TO + GO = OUT.**

Crypt - arithmetic Problems

- **Unique Letter-Digit Mapping**:
  - Each letter represents a unique digit ranging from 0 to 9.
  - A digit cannot correspond to more than one letter.
- **Rules for Leading Digits**:
  - Numbers cannot begin with the digit zero (e.g., 0123 is incorrect, while 123 is correct).
- **Carry in Addition**:
  - In addition problems, any carryover is strictly limited to the value of 1.

- There must be only one solution .

Example TO + GO = OUT

Step 1 Identify the variables

- The equation has letters: T, O, G, U.
- Each letter represents a unique digit (0-9).
- The digits must satisfy the equation, and no number can start with 0.

Step 2 Analyse the equation structure

- **TO** and **GO** are two-digit numbers.
- **OUT** is a three-digit number.
- T + G is generating O is carry
  - value of O is 1

```
  T O
+ G O
-----
O U T
```

Example TO + GO = OUT (continued)

```
  T O
+ G O
-----
O U T
```

Step 3 Determine the Value of T

- Substitution of value of O is 1
- The digits must satisfy the equation, and no number can start with 0.

Step 4 Analyze G and U.

```
  T 1
+ G 1
-----
1 U T
```

```
  2 1
+ G 1
-----
1 U 2
```

Example TO + GO = OUT (continued)

```
  2 1
+ G 1
-----
1 U 2
```

Step 4 Analyse G and U.

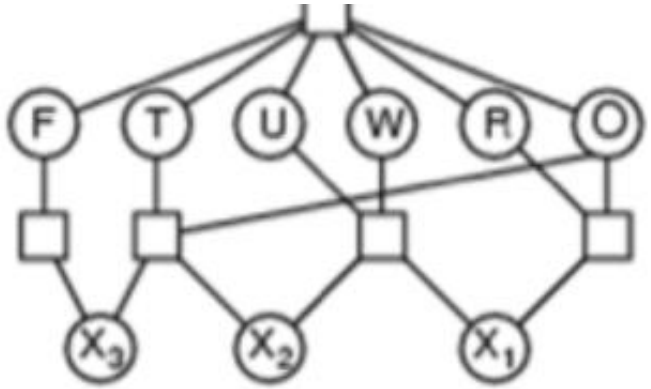
Now, 2 + G > 10 (as its resulting a carry 1 on next)

- Now, possible values of G to get 1 carry at next step is – {G = 8 or 9}
  - If G is 9 then – U = 2 + 9 = 11
  - So value of U becomes 1 and 1 goes to carry
  - Now, value of O is already 1 so U value can not be 1 also. **(game rule)**
- Now, the value of G has to be 8, lets verify this also
  - U = 2 + 8 = 10
  - So value of U becomes 0, there is no conflict.
- So final values are – T = 2, O = 1, G = 8, U = 0

Example

- Variables: $F,T,U,W,R,O \;\; X_1 X_2 X_3$
- Domains: $\{0,1,2,3,4,5,6,7,8,9\}$
- Constraints: $Alldiff\,(F,T,U,W,R,O)$
  - $O + O = R + 10 \cdot X_1$
  - $X_1 + W + W = U + 10 \cdot X_2$
  - $X_2 + T + T = O + 10 \cdot X_3$
  - $X_3 = F,\; T \neq 0,\; F \neq 0$

$$\begin{array}{r} T\;W\;O \\ +\;T\;W\;O \\ \hline F\;O\;U\;R \end{array}$$



Exercise

$$\begin{array}{r} S\ E\ N\ D \\ +\ \ M\ O\ R\ E \\ \hline =\ M\ O\ N\ E\ Y \end{array} \qquad \begin{array}{r} E\ A\ T \\ +\ \ T\ H\ A\ T \\ \hline =\ A\ P\ P\ L\ E \end{array} \qquad \begin{array}{r} B\ A\ S\ E \\ +\ \ B\ A\ L\ L \\ \hline =\ G\ A\ M\ E\ S \end{array}$$

Real-world CSPs

- Assignment problems
  - e.g., who teaches what class
- Timetabling problems
  - e.g., which class is offered when and where?
- Transportation scheduling
- Factory scheduling

- Notice that many real-world problems involve real-valued variables

Backtracking search

- Variable assignments are commutative}, i.e.,

[WA = red then NT = green] same as [NT = green then WA = red]

- Only need to consider assignments to a single variable at each node
  → b = d and there are d^n leaves
- Depth-first search for CSPs with single-variable assignments is called backtracking search
- Backtracking search is the basic uninformed algorithm for CSPs
- Can solve *n*-queens for $n \approx 25$

Backtracking search

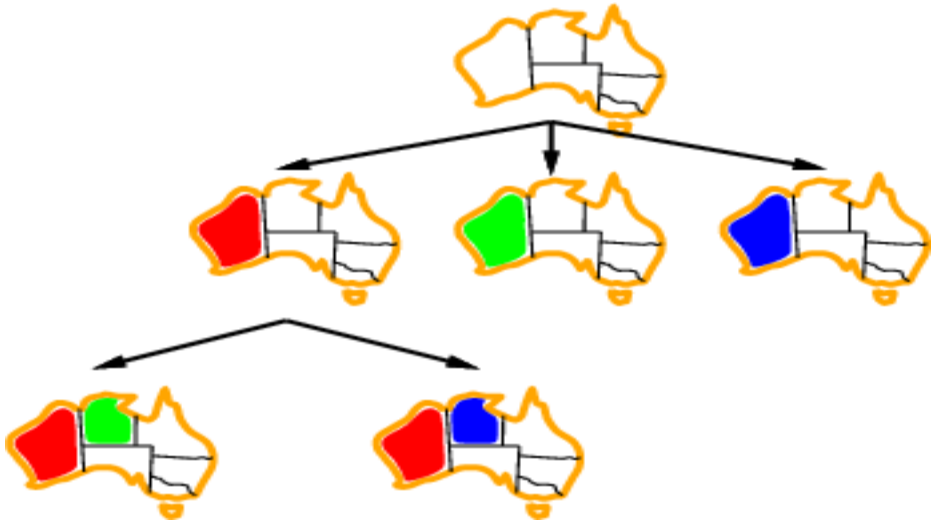
```
function BACKTRACKING-SEARCH( csp) returns a solution, or failure
    return RECURSIVE-BACKTRACKING({}, csp)

function RECURSIVE-BACKTRACKING( assignment, csp) returns a solution, or
failure
    if assignment is complete then return assignment
    var ← SELECT-UNASSIGNED-VARIABLE( Variables[csp], assignment, csp)
    for each value in ORDER-DOMAIN-VALUES(var, assignment, csp) do
        if value is consistent with assignment according to Constraints[csp] then
            add { var = value } to assignment
            result ← RECURSIVE-BACKTRACKING(assignment, csp)
            if result ≠ failue then return result
            remove { var = value } from assignment
    return failure
```

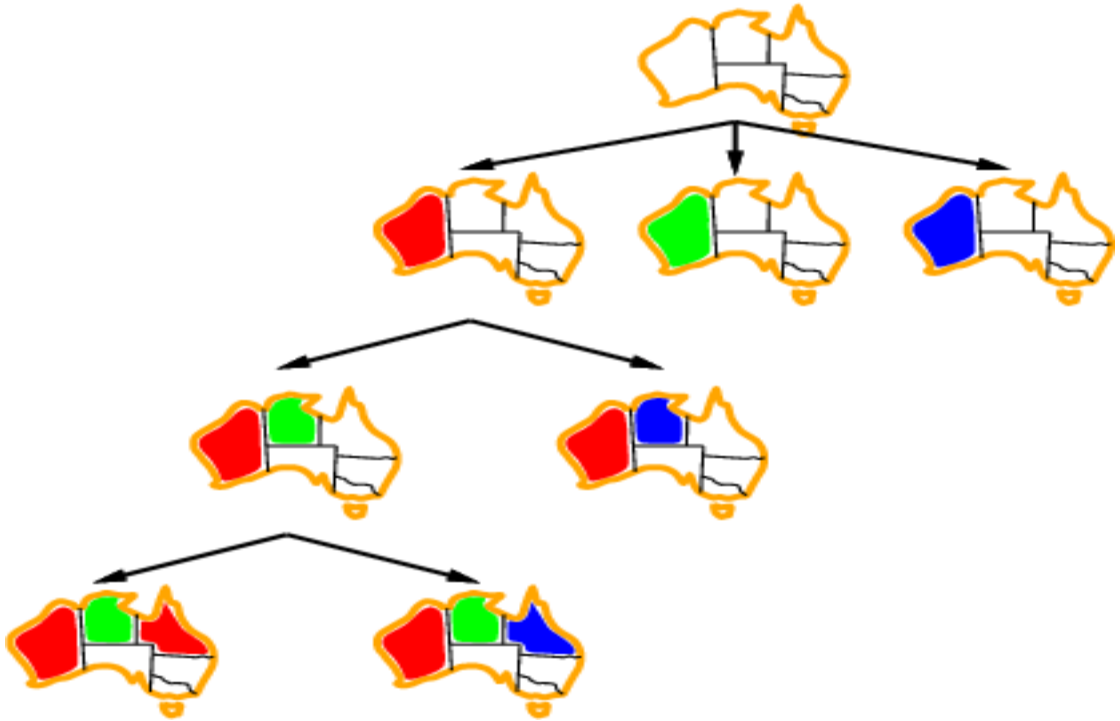
Backtracking example

Backtracking example

Backtracking example



Backtracking example



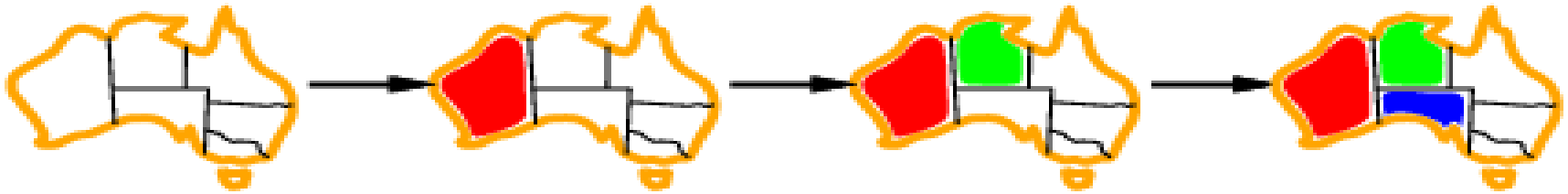
Improving backtracking efficiency

- General-purpose methods can give huge gains in speed:
-
  - Which variable should be assigned next?
  - In what order should its values be tried?
  - Can we detect inevitable failure early?

Most constrained variable

- Most constrained variable:

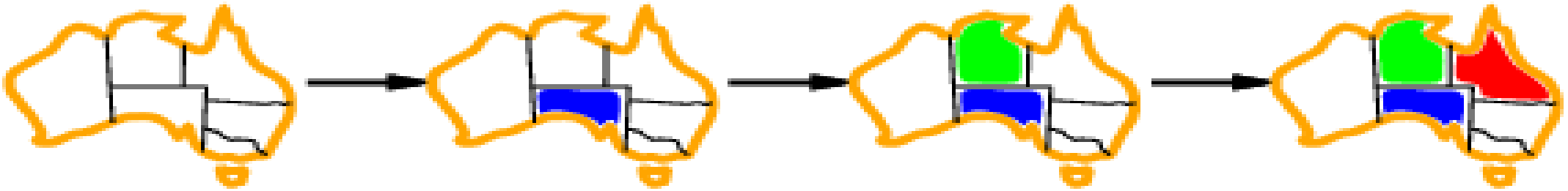
  choose the variable with the fewest legal values



- a.k.a. minimum remaining values (MRV) heuristic
-

Most constraining variable

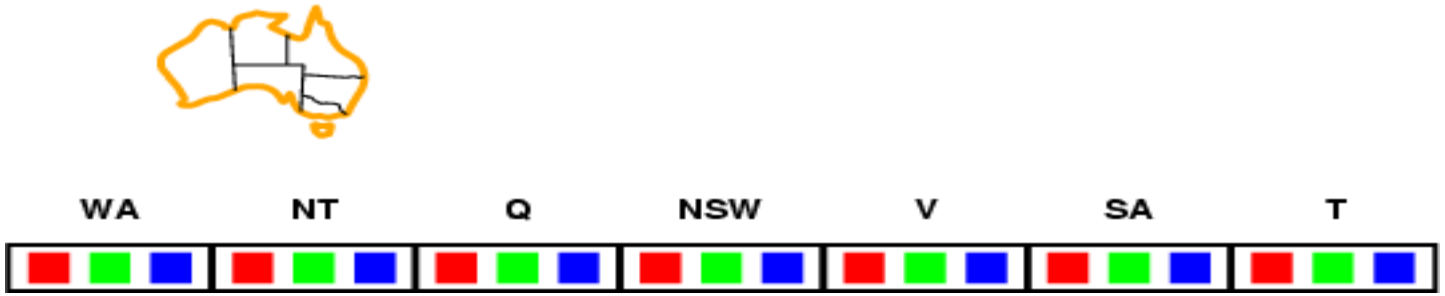
- Tie-breaker among most constrained variables
- Most constraining variable:
-
  - choose the variable with the most constraints on remaining variables
  -



Least constraining value

- Given a variable, choose the least constraining value:
-
  - the one that rules out the fewest values in the remaining variables
  -
- Cc

Forward checking

- Idea:
  - Keep track of remaining legal values for unassigned variables
  - Terminate search when any variable has no legal values
  - 

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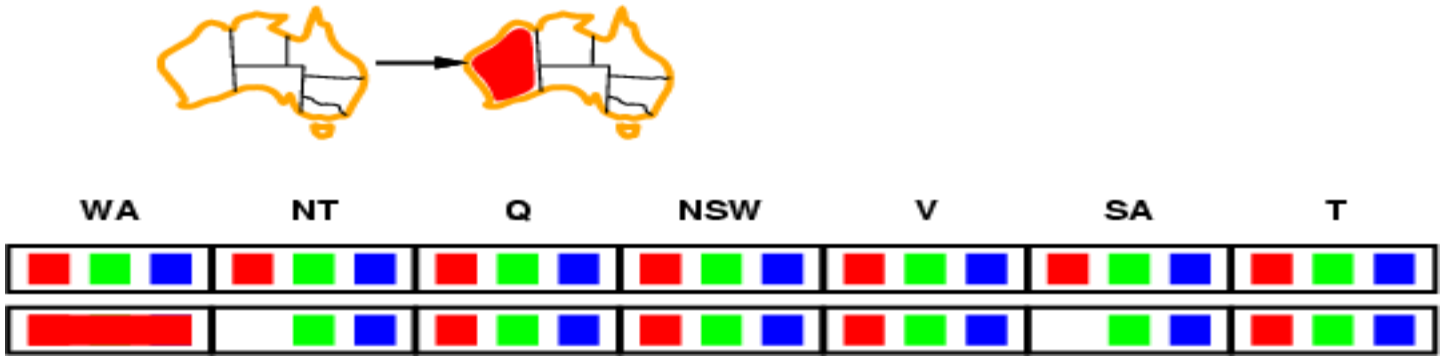
  - 

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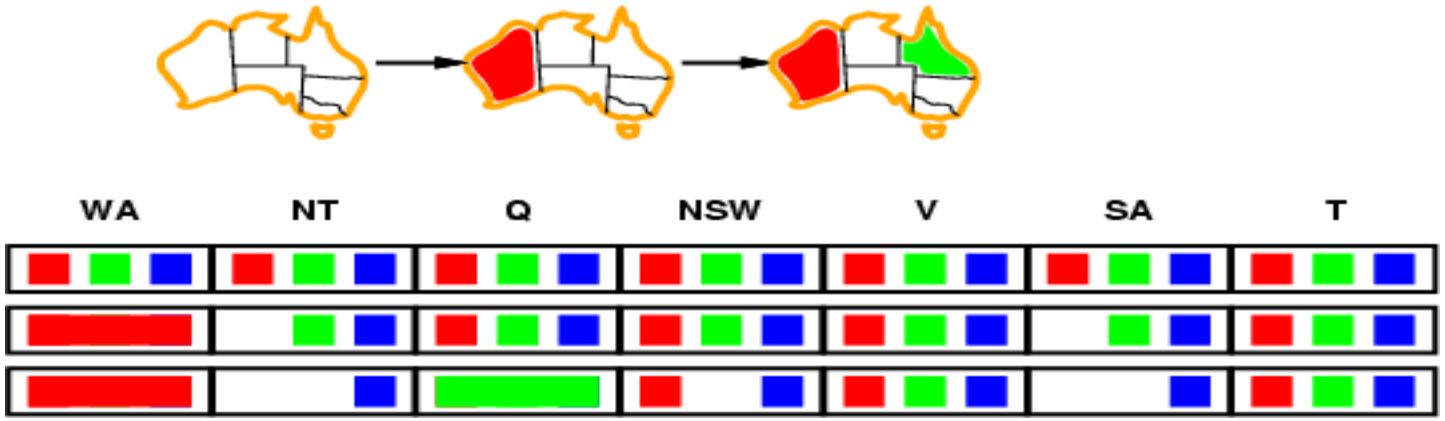
  - 

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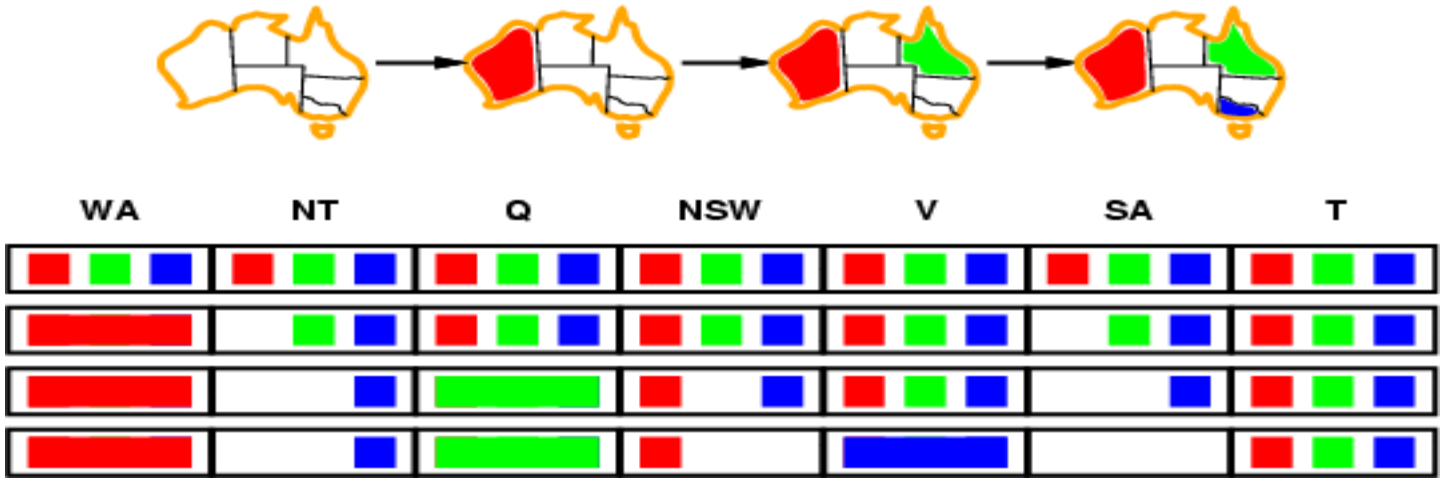
  - 

Constraint propagation

- Forward checking propagates information from assigned to unassigned variables, but doesn't provide early detection for all failures:
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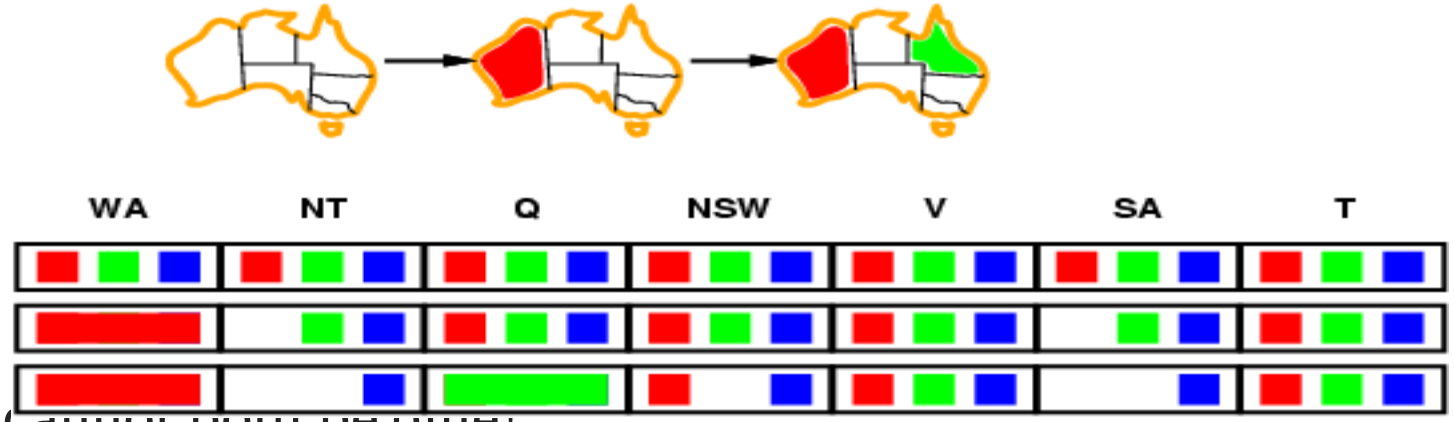
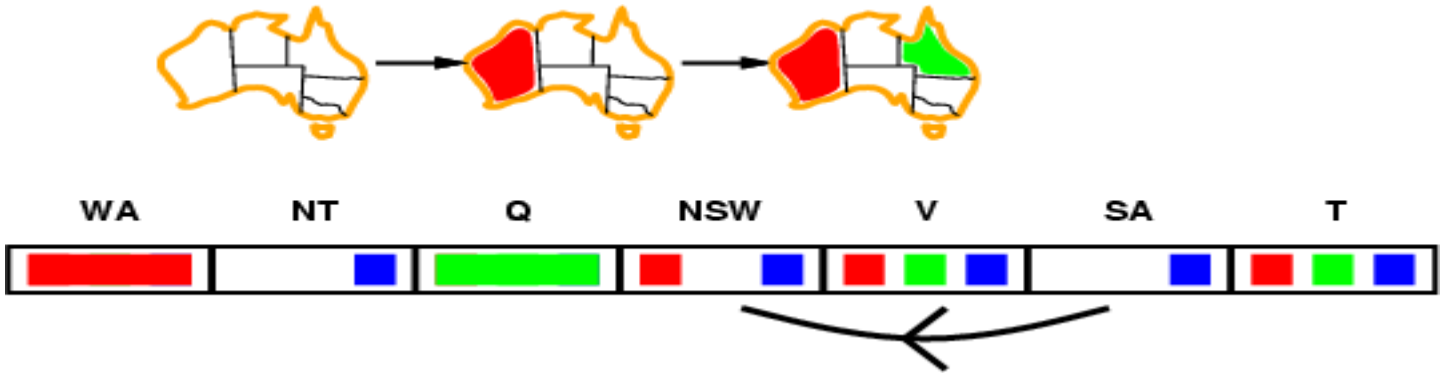


- NT and SA cannot both be blue!
- Constraint propagation repeatedly enforces constraints locally
-

Arc consistency

- Simplest form of propagation makes each arc consistent
- $X \rightarrow Y$ is consistent iff
-

  for every value x of X there is some allowed y



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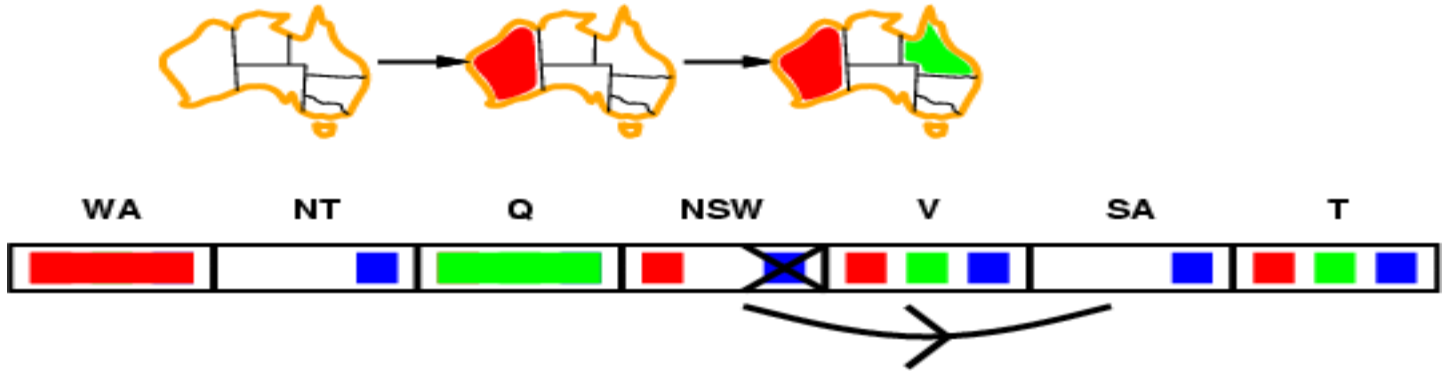

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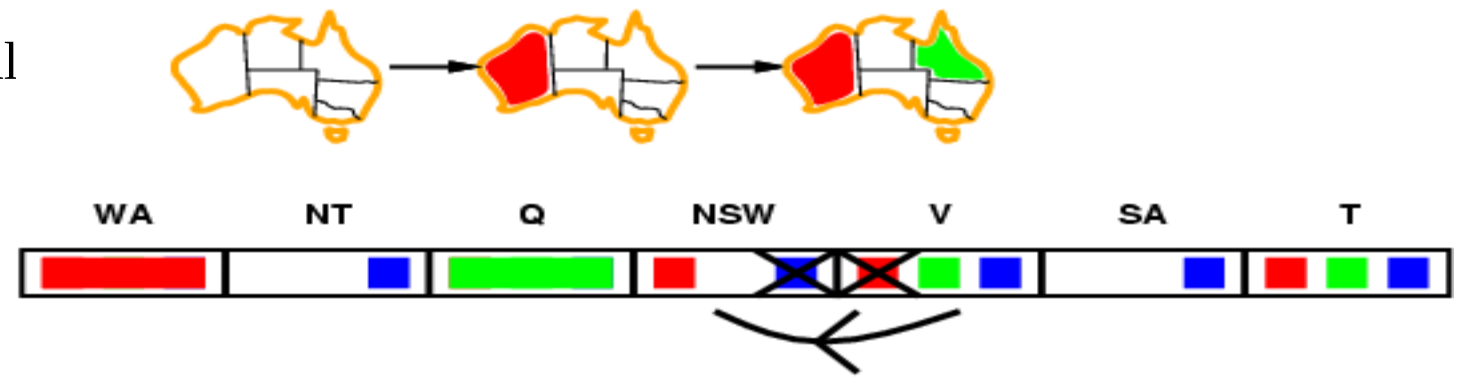

- If X loses a value, neighbors of X need to be rechecked
-

Arc consistency

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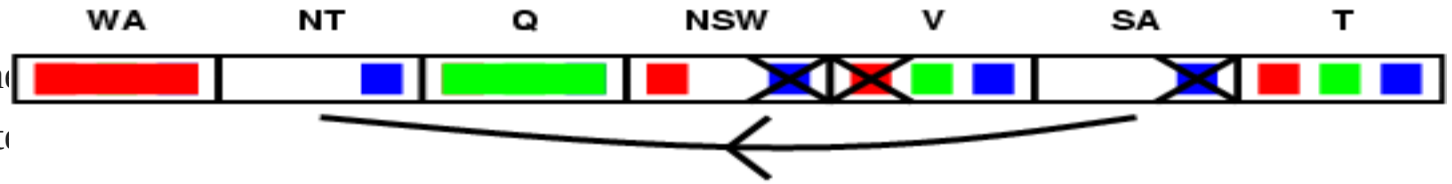
- $X \rightarrow Y$ is consistent iff
- for every value x of X there is some allowed y



- If X loses a value, n
- Arc consistency det
-
- Can be run as a preprocessor or after each assignment
-

Arc consistency algorithm AC-3

```
function AC-3( csp) returns the CSP, possibly with reduced domains
   inputs: csp, a binary CSP with variables {X1, X2, ..., Xn}
   local variables: queue, a queue of arcs, initially all the arcs in csp

   while queue is not empty do
      (Xi, Xj) ← REMOVE-FIRST(queue)
      if RM-INCONSISTENT-VALUES(Xi, Xj) then
         for each Xk in NEIGHBORS[Xi] do
            add (Xk, Xi) to queue
```

```
function RM-INCONSISTENT-VALUES( Xi, Xj) returns true iff remove a value
   removed ← false
   for each x in DOMAIN[Xi] do
      if no value y in DOMAIN[Xj] allows (x,y) to satisfy constraint(Xi, Xj)
         then delete x from DOMAIN[Xi];   removed ← true
   return removed
```

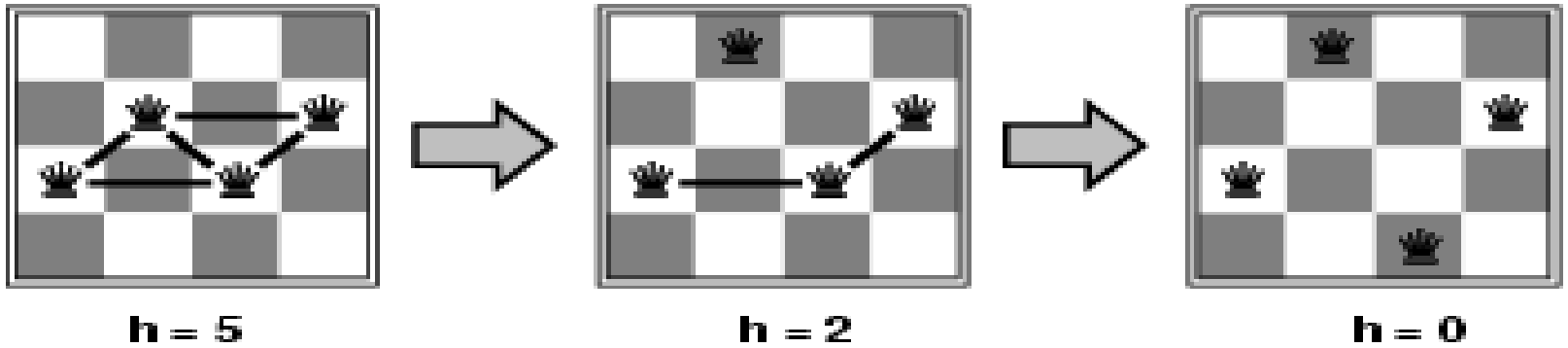
- Time complexity: $O(n^2d^3)$
-

Local search for CSPs

- Hill-climbing, simulated annealing typically work with "complete" states, i.e., all variables assigned
- To apply to CSPs:
  - allow states with unsatisfied constraints
  - operators reassign variable values
- Variable selection: randomly select any conflicted variable
- Value selection by min-conflicts heuristic:
  - choose value that violates the fewest constraints
  - i.e., hill-climb with $h(n)$ = total number of violated constraints

Example: 4-Queens

- States: 4 queens in 4 columns ($4^4 = 256$ states)
- Actions: move queen in column
- Goal test: no attacks
- Evaluation: $h(n)$ = number of attacks



- Given random initial state, can solve n-queens in almost constant time for arbitrary n with high probability (e.g., n = 10,000,000)

Summary

- CSPs are a special kind of problem:
  - states defined by values of a fixed set of variables
  - goal test defined by constraints on variable values

- Backtracking = depth-first search with one variable assigned per node
- Variable ordering and value selection heuristics help significantly
- Forward checking prevents assignments that guarantee later failure
- Constraint propagation (e.g., arc consistency) does additional work to constrain values and detect inconsistencies
- Iterative min-conflicts is usually effective in practice

THANK YOU